Maths for Computing — Phase 3: Algorithms, Relations, Functions and Recursion

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What this phase covers

This phase is Lecture 3 and Seminar 3. It unlocks mock **Questions 2, 4, 12, 19, 20 and 21**, plus revision Tasks 6, 7, 8, 10 and 11.

You need from earlier phases: solving equations and rearranging formulas, quadratics, and the coordinate plane (Phase 0); ordered pairs and Cartesian products (Phase 2).

1. Algorithms

An **algorithm** is a **finite** (it must end) sequence of **well-defined** (clear, unambiguous) instructions for completing a task. A recipe is an everyday algorithm.

Why algorithms matter: **efficiency** (faster), **problem-solving** (a systematic method) and **resource optimisation** (using memory and processor time well).

Two basic families:

- **Searching algorithms** find an item in a collection of data: linear search, binary search.
- **Sorting algorithms** put data in order: bubble sort, quick sort.

Algorithms are often drawn as **flowcharts**: boxes for steps, diamonds for yes/no decisions, and arrows for the order.

Linear search

Check each element **one by one from the start** until you find the target or run out.

Formally: given $S = \{a_1, a_2, ..., a_n\}$ and a target d, for i = 1 to n: if $a_i = d$, return i (its position). If nothing matches, return "Not Found".

Example (lecture): list 12, 34, 25, 6, 17. Search for 25.

Step	Element checked	Equal to 25?
1	12	No

Step	Element checked	Equal to 25?
2	34	No
3	25	Yes — found at position 3

It took **3 comparisons**. Searching for 17 would take 5. Searching for 99 (not there) takes 5 and returns "Not Found".

Linear search works on **unsorted** lists but is slow on big lists.

Binary search

Binary search works **only on a sorted list**. It checks the middle element, then throws away the half that can't contain the target. Each step halves the search area.

The variables:

- **L** = left bound (first position still being searched). Starts at 1.
- **R** = right bound (last position still being searched). Starts at n, the list length.
- **M** = the middle position = (L + R) / 2, **rounded down** to a whole number.

Repeat while $L \leqslant R$:

1. Compute $M = \lfloor (L + R)/2 \rfloor$. (The $\lfloor\ \rfloor$ brackets mean "round down", called the **floor**.)
2. If $a_m = d$, you've found it: return M.
3. If $a_m < d$, the target must be to the right: set **L = M + 1**.
4. If $a_m > d$, the target must be to the left: set **R = M − 1**.

If L becomes bigger than R, the target is not in the list: return "Not Found".

Example (lecture): sorted list 3, 7, 11, 23, 27, 31, 37, 43, 47 (positions 1 to 9). Search for 37.

Step	L	R	M	a_m	Comparison	Action
1	1	9	5	27	27 < 37	L = 6
2	6	9	7	37	37 = 37	Found at position 7

Example 2: same list, search for 11.

Step	L	R	M	a_m	Comparison	Action
1	1	9	5	27	27 > 11	R = 4

Step	L	R	M	a_m	Comparison	Action
2	1	4	2 (2.5 rounded down)	7	7 < 11	L = 3
3	3	4	3 (3.5 rounded down)	11	equal	Found at position 3

Example 3: same list, search for 30 (not present).

Step	L	R	M	a_m	Action
1	1	9	5	27	27 < 30, so L = 6
2	6	9	7	37	37 > 30, so R = 6
3	6	6	6	31	31 > 30, so R = 5
4	L = 6 > R = 5				Stop: Not Found

Bubble sort

Bubble sort walks through the list comparing **neighbours**. If a pair is in the wrong order (left bigger than right), **swap** them. Each full walk-through is a **pass**. After each pass, the largest remaining number has "bubbled" to the end.

Formally: for i = 1 to n, for j = 1 to n − i, if $a_j > a_{j+1}$, swap them. The n − i means each pass can stop one place earlier, since the end is already sorted.

Example (lecture): sort 29, 10, 14, 37, 13.

Pass 1 (4 comparisons):

Compare	Swap?	List after
29, 10	Yes	10, 29, 14, 37, 13
29, 14	Yes	10, 14, 29, 37, 13
29, 37	No	10, 14, 29, 37, 13
37, 13	Yes	10, 14, 29, 13, **37**

Pass 2 (3 comparisons): (10, 14) no; (14, 29) no; (29, 13) yes → 10, 14, 13, **29, 37**.

Pass 3 (2 comparisons): (10, 14) no; (14, 13) yes → 10, 13, **14, 29, 37**.

Pass 4 (1 comparison): (10, 13) no. Final: **10, 13, 14, 29, 37**.

Quick sort (just know the idea): pick an element, split the list into smaller and larger parts, and sort each part the same way. Much faster than bubble sort on big lists.

Representing data (lecture summary)

Category	Examples	Used for
Tabular	Frequency tables, contingency tables	Counting outcomes; relationships between categories
Graphical	Bar graphs, histograms, scatter plots, line graphs, pie charts	Visualising frequencies, trends and parts of a whole
Algebraic	Equations, inequalities, functions	Describing relationships between variables

Mini tasks 1

1. How many comparisons does linear search need to find 6 in 12, 34, 25, 6, 17?
2. Binary search for 43 in 3, 7, 11, 23, 27, 31, 37, 43, 47. List M and a_m at each step.
3. Write the list after the **first pass** of bubble sort on 5, 1, 4, 2, 8.
4. Why can't you use binary search on 12, 34, 25, 6, 17?

2. Relations

A **relation** R from set A to set B is **any subset of the Cartesian product A × B**. So it's a set of ordered pairs (a, b), with a from A and b from B. We write $R \subseteq A \times B$.

A relation says which elements of A are "connected" to which elements of B. There are no other rules: an element can be connected to many, one or none.

Example (lecture): A = {1, 2}, B = {x, y}. A × B = {(1, x), (1, y), (2, x), (2, y)}. One possible relation is R = {(1, x), (1, y)}. Another is R = { } (the empty relation). A × B itself is also a relation.

Your course sometimes writes pairs with angle brackets: ⟨a, b⟩ means the same as (a, b).

Ways to show a relation

- **A set of pairs:** {(1, x), (1, y)}

- **An arrow diagram:** two ovals, A on the left and B on the right, with an arrow from each a to its b
- **A table** with A down the side and B across the top, ticking connected pairs
- **Points on a graph** (Cartesian coordinate system)

In computing, relations model networks, database links and social connections, where one item may link to several others.

3. Functions

A **function** f from A to B, written **f: A → B**, is a special relation where **every element of A is paired with exactly one element of B**.

In plain words: **every input gives one and only one output.**

Every function is a relation, but not every relation is a function.

The vocabulary

- **Domain:** the set of allowed inputs (A).
- **Co-domain:** the set of **possible** outputs (B). It's the target set, whether or not every element is used.
- **Range** (or image): the outputs that **actually** occur. The range is always a subset of the co-domain.
- **Rule:** how each input is turned into its output, such as f(x) = x + 3.
- **f(x)** is read "f of x": the output when the input is x. If f(x) = x + 3, then f(1) = 4 and f(14) = 17.

The two-part test (memorise this)

A relation from A to B is a **function with domain A** if and only if:

1. **Every** element of A appears as a first item. (No input left out.)
2. **No** element of A appears as a first item **twice** with different outputs. (No input gets two outputs.)

Things that are **allowed**:

- Two different inputs going to the **same** output, like Mary → Programmer and Chris → Programmer.
- Some elements of B not being used at all.

Each output must also actually be in B.

Worked examples (lecture Activity 1)

Relation	Function?	Reason
R = {(1, 2), (2, 3), (3, 4), (4, 5)}	Yes	Each input 1, 2, 3, 4 has one output
S = {(a, 1), (b, 2), (a, 3), (c, 1)}	No	a has two outputs (1 and 3)
T = {(0, 0), (1, 2), (2, 4), (1, 3)}	No	1 has two outputs (2 and 3)
U = {(x, y) | y = 2x, x ∈ {−2, −1, 0, 1, 2}}	Yes	The formula gives exactly one y for each x
V = {(a, b) | a ∈ {1, 2, 3}, b ∈ {4, 5, 6}}	No	This is a full Cartesian product: each a goes to 4, 5 and 6

Seminar 3 Task 1

A = {Mary, John, Peter, Chris}, B = {Accountant, Lawyer, Programmer, Lecturer}.

Relation	Function?	Reason
1. Mary→Accountant, John→Lawyer, Peter→Programmer, Chris→Lecturer	Yes	Everyone has exactly one job
2. …Peter→Programmer, Chris→Programmer	Yes	Sharing an output is allowed
3. Mary, John and Peter only	**No**	Chris is missing
4. Everyone → Programmer	Yes	One output each (even if it's the same one)
5. Peter→Lecturer and Peter→Programmer	**No**	Peter has two outputs

Mock Question 4 solved

R = {⟨a, 0⟩, ⟨b, 1⟩, ⟨c, 1⟩}. In which situation is R a function with domain A and co-domain B?

- (A) A = {a, b, c, d}: **no**, d has no output.
- (B) B = {0}: **no**, the outputs 1 aren't in B.

- (C) B = { }: **no**, no outputs are in B.
- (D) A = {a, b, c}, B = {1, 0}: every input once, every output in B. **Yes. Answer D.**

Mock Question 12 solved

A = {p, q}, B = {0, 1}.

- (A) {⟨p, q⟩, ⟨0, 1⟩}: q isn't in B, and 0 isn't in A. Not a function from A to B.
- (B) {⟨p, 0⟩, ⟨q, 1⟩, ⟨q, 0⟩}: q has two outputs. No.
- (C) {⟨p, 1⟩, ⟨q, 0⟩, ⟨p, 0⟩}: p has two outputs. No.
- (D) {⟨p, 1⟩, ⟨q, 0⟩}: p and q each have exactly one output. **Yes. Answer D.**

Graphs: the vertical line test

For a graph, draw imaginary vertical lines. If any vertical line hits the graph **more than once**, one x has two y values, so it's **not a function**. A circle fails; a straight slanted line or a parabola passes.

Mini tasks 3

A = {1, 2, 3}, B = {x, y}. Which are functions from A to B?

1. {(1, x), (2, y), (3, x)}
2. {(1, x), (2, y)}
3. {(1, x), (1, y), (2, x), (3, y)}
4. {(1, y), (2, y), (3, y)}
5. For the function f(x) = 2x − 1 with domain {0, 1, 2}, what is the range?

4. Types of functions

You should recognise each of these by its formula and by the shape of its graph (Seminar 3 Task 4).

Type	Formula	Graph shape	Inverse
Linear	f(x) = mx + b	Straight line; grows at a constant rate	$f^{-1}(x) = (x - b)/m$
Identity	u(x) = x	Straight line through (0, 0) at 45°	Itself

Type	Formula	Graph shape	Inverse
Squared (quadratic)	$h(x) = x^2$	U-shaped **parabola**, lowest point (vertex) at (0, 0), symmetric about the y-axis	Only if the domain is restricted: $\sqrt{x}$ for $x \geq 0$
Natural exponential	$f(x) = e^x$ (or $a \cdot e^{bx}$)	Starts almost flat near 0 on the left, then shoots up rapidly; always above the x-axis; passes (0, 1)	$\ln(x)$
Logarithmic	$f(x) = \log_b(x)$, e.g. $\ln(x)$	Only exists for $x > 0$; passes (1, 0); rises slowly; plunges down near the y-axis	b^x (exponential)
Reciprocal	$g(x) = 1/x$	Two separate curves, in the top-right and bottom-left; never touches either axis; undefined at $x = 0$	Itself

Notes:

- **e ≈ 2.71828** is Euler's number. Exponential functions model compound interest and population growth.
- **ln(x)** is the natural logarithm, log base e. It is the inverse of e^x: $\ln(e^x) = x$.
- A **logarithm** answers "what power?": $\log_2(8) = 3$ because $2^3 = 8$.
- An **asymptote** is a line the graph gets closer and closer to but never touches. e^x has the x-axis as an asymptote; $\ln(x)$ has the y-axis.

The graphs of a function and its inverse are **mirror images in the line y = x**. That's why e^x and $\ln(x)$ look like reflections of each other.

5. Linear functions and straight lines

This section matters most for the exam: mock Questions 2 and 20, and revision Tasks 6 and 10.

Slope-intercept form

```
y = mx + b
```

- **m is the slope** (or gradient): how steep the line is, and which way it goes.
- **b is the y-intercept**: where the line crosses the y-axis, at the point (0, b).

Writing a line in this form is called **slope-intercept form**. Some books write y = mx + c; it's the same thing.

To read m and b, the equation must be in the form y = (number)x + (number). Example: y = ½x − ¾ has m = ½, b = −¾. And 3(2x − 5) + 4 must be expanded first: 6x − 11, so m = 6, b = −11.

What the slope tells you

Slope	Line	Example
Positive (m > 0)	Rises from left to right	y = 2x
Negative (m < 0)	Falls from left to right	y = −2x
Zero (m = 0)	Horizontal	y = 3
Undefined	Vertical	x = 3 (not a function)

A bigger |m| means a steeper line. m = 5 is steeper than m = ½.

The slope means: **for every 1 step right, the line goes m steps up.** If m = ½, it goes up half a unit for each unit right, or up 1 for every 2 right.

Changing b slides the whole line up or down without changing its steepness.

Slope from two points

```
m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}
```

Δ (delta) means "change in". This is "rise over run". Subtract the coordinates in the **same order** on top and bottom.

Example (lecture): points (1, 2) and (4, 8). m = (8 − 2)/(4 − 1) = 6/3 = **2**.

Example with negatives (Seminar 3 Task 3): (6, −1) and (−2, 7). m = (7 − (−1))/(−2 − 6) = 8/(−8) = **−1**.

Finding the equation of a line through two points

1. Find m with the formula.

2. Put m and one point into y = mx + b and solve for b.
3. Write y = mx + b.

Example (lecture): (1, 2) and (4, 8). m = 2. Using (1, 2): 2 = 2 × 1 + b, so b = 0. Line: **y = 2x**.

Example (Seminar Task 3): m = −1. Using (6, −1): −1 = −1 × 6 + b = −6 + b, so b = 5. Line: **y = −x + 5**. Check with (−2, 7): −(−2) + 5 = 7. ✓

Example (revision Task 10A): D(2, 3), E(5, 9). m = 6/3 = 2. 3 = 2 × 2 + b, so b = −1. Line: **y = 2x − 1**.

Alternative: point-slope form. $y - y_1 = m(x - x_1)$. With D(2, 3): y − 3 = 2(x − 2) → y = 2x − 1. Same answer.

Where a line crosses the axes

- **y-intercept:** put **x = 0**. You get y = b. The point is **(0, b)**.
- **x-intercept:** put **y = 0** and solve 0 = mx + b, giving x = −b/m. The point is **(−b/m, 0)**.

Full function analysis (Seminar 3 Task 2): h(x) = ½x − ¾

1. **Type:** linear (form mx + b, with m = ½ and b = −¾).
2. **Table of values:**

x	−2	−1	0	1	2
h(x)	−1.75	−1.25	−0.75	−0.25	0.25

Ordered pairs: (−2, −1.75), (−1, −1.25), (0, −0.75), (1, −0.25), (2, 0.25).

3. **y-intercept:** h(0) = −¾, so **(0, −0.75)**.
4. **x-intercept:** 0 = ½x − ¾ → ½x = ¾ → x = 1.5, so **(1.5, 0)**.

Picking the right graph without a calculator (mock Question 20)

Which graph is h(x) = ½x − ¾?

Use three checks:

1. **Direction:** m = ½ > 0, so it **rises**. Rule out D, which falls.
2. **y-intercept:** −¾, just **below** the origin. Rule out C, which passes through (0, 0). Rule out A, which also goes through the origin.

3. **Steepness and x-intercept:** m = ½ is gentle (up 1 for every 2 right), and the line crosses the x-axis at x = 1.5, just right of 0. Graph A is far too steep (slope 1).

Graph **B** crosses the x-axis slightly right of 0, rises gently and sits slightly below the origin at x = 0. **Answer B.**

General method: compute the y-intercept and the x-intercept, then find the graph passing through both.

Mini tasks 5

1. Give the slope and y-intercept of y = −3x + 7.
2. Find the slope through (−1, 4) and (3, −4).
3. Find the equation of the line through (1, 3) and (3, 7).
4. Find both intercepts of g(x) = (3/4)x − 5/2.
5. Does y = −x + 2 rise or fall? Where does it cross the y-axis?

6. Parallel and perpendicular lines

Parallel: same slope

Two lines are **parallel** (they never meet) when they have the **same slope** but **different y-intercepts**. If the y-intercepts were also the same, they would be the same line.

- f(x) = 3x + 2 and g(x) = 3x − 4: slopes 3 and 3. **Parallel.**
- f(x) = 4x − 5 and g(x) = −4x + 3: slopes 4 and −4. **Not parallel.**
- f(x) = 3(2x − 5) + 4 = 6x − 11 and g(x) = 6x − 2: slopes 6 and 6. **Parallel.** (Expand first!)

Why they never meet: same steepness, different starting heights, so the gap between them never changes.

Making a parallel line (Seminar Task 6b): parallel to p(x) = −2x + 6 with y-intercepts 4 and −7: **y = −2x + 4** and **y = −2x − 7**. Keep m, change b.

Perpendicular: slopes multiply to −1

Two lines are **perpendicular** (they cross at a right angle, 90°) when:

```
m_1 \times m_2 = -1 \quad\Longleftrightarrow\quad m_2 = -\frac{1}{m_1}
```

So the perpendicular slope is the **negative reciprocal**: flip the fraction upside down, and change the sign.

Slope m_1	Flip it	Change sign: m_2
2	1/2	−1/2
−1/2	−2	2
3	1/3	−1/3
9/5	5/9	−5/9
−4	−1/4	1/4

Classifying pairs (revision Task 10B)

- 2x + 3 and 2x − 5: equal slopes → **parallel**.
- −½x + 4 and 2x − 1: (−½) × 2 = −1 → **perpendicular**.
- 3x + 2 and −⅓x + 1: 3 × (−⅓) = −1 → **perpendicular**.
- 4x − 7 and ¼x + 2: not equal, and 4 × ¼ = +1, not −1 → **neither**. (The sign matters!)

Mock Question 2 fully solved

Find the line through A(4, −3) that is perpendicular to the line through B(2, 5) and C(−3, −4).

Step 1: slope of BC.

```
m_{BC} = \frac{-4 - 5}{-3 - 2} = \frac{-9}{-5} = \frac{9}{5}
```

Step 2: perpendicular slope. Flip 9/5 to 5/9, change the sign: **m = −5/9**.

Step 3: find b using A(4, −3).

- −3 = (−5/9) × 4 + b
- −3 = −20/9 + b
- b = −3 + 20/9 = −27/9 + 20/9 = **−7/9**

Step 4: y = −(5/9)x − 7/9. **Answer D.**

Exam shortcut: after Step 2, only options with slope −5/9 can be right (C and D). Then test point A in each: for D, −(5/9)(4) − 7/9 = −27/9 = −3 ✓. For C, −20/9 + 1 ≠ −3 ✗.

Mini tasks 6

1. Are $y = -3x + 2$ and $y = (1/3)x - 5$ parallel, perpendicular or neither?
2. What slope is perpendicular to $y = (2/7)x + 1$?
3. Find the line perpendicular to $y = 4x - 1$ that passes through $(4, 2)$.
4. Find the line parallel to $y = 5x + 2$ that passes through $(1, 1)$.

7. Inverse functions

The idea

The **inverse** of f, written $\mathbf{f^{-1}}$, **undoes** f. If f turns 2 into 7, then f^{-1} turns 7 back into 2.

Formally, $f: A \to B$ has an inverse $f^{-1}: B \to A$ when, for all x and y:

$$f(f^{-1}(y)) = y \qquad\text{and}\qquad f^{-1}(f(x)) = x$$

Important: **$\mathbf{f^{-1}(x)}$ does not mean $\mathbf{1/f(x)}$.** The −1 here is a label, not a power.

When does an inverse exist? Bijective functions

A function has an inverse only if it is **bijective**, meaning both:

- **One-to-one (injective):** no two different inputs give the same output. (Otherwise, going backwards, one output would need to return to two inputs.)
- **Onto (surjective):** every element of the co-domain B is an output of some input. (Otherwise some elements of B would have nowhere to go back to.)

Horizontal line test: if any horizontal line hits the graph twice, the function is not one-to-one, so no inverse.

Example: $f(x) = x^2$ on all real numbers fails: $f(2) = f(-2) = 4$. So x^2 has no inverse over $\mathbb{R}$. But if we **restrict the domain** to $x \geq 0$, it becomes one-to-one, and the inverse is $\sqrt{x}$. (If the domain is $x \leq 0$, the inverse is $-\sqrt{x}$.)

Linear functions with $m \neq 0$ always have inverses.

The method: swap x and y, then solve for y

1. Write $y = f(x)$.
2. **Swap** x and y.
3. **Solve** for y (rearrange so y is alone).

4. Write $f^{-1}(x)$ = (the result).

Example 1 (lecture): $f(x) = 2x + 3$

1. $y = 2x + 3$
2. $x = 2y + 3$
3. $x - 3 = 2y \rightarrow y = (x - 3)/2$
4. **$f^{-1}(x) = (x - 3)/2$**

Check: $f(5) = 13$, and $f^{-1}(13) = 10/2 = 5$. ✓

General linear rule: if $f(x) = mx + b$, then $f^{-1}(x) = (x - b)/m$.

Example 2 (Seminar Task 8): $h(x) = (3x - 4)/2$

1. $x = (3y - 4)/2$
2. Multiply by 2: $2x = 3y - 4$
3. Add 4: $2x + 4 = 3y$
4. Divide by 3: **$h^{-1}(x) = (2x + 4)/3$**

Algebraic check: $h(h^{-1}(x)) = (3 \times (2x + 4)/3 - 4)/2 = (2x + 4 - 4)/2 = 2x/2 = x$. ✓

Example 3 (revision Task 6): $g(x) = (3/4)x - 5/2$

1. $x = (3/4)y - 5/2$
2. Add 5/2: $x + 5/2 = (3/4)y$
3. Multiply by 4/3: $y = (4/3)x + (4/3)(5/2) = (4/3)x + 10/3$
4. **$g^{-1}(x) = (4/3)x + 10/3$**

Example 4 (mock Question 21): $f(x) = x^2 + 2$, domain $x \geq 0$

1. $x = y^2 + 2$
2. Subtract 2: $x - 2 = y^2$
3. Square root (positive, since the original domain was $x \geq 0$): $y = \sqrt{(x - 2)}$
4. **$f^{-1}(x) = \sqrt{(x - 2)}$. Answer A.**

Check: $f(3) = 11$, and $\sqrt{(11 - 2)} = \sqrt{9} = 3$. ✓ Option C, $\sqrt{(x + 2)}$, would give $\sqrt{13}$. ✗

Quick-check trick for multiple choice: pick an easy input, compute f of it, then plug that output into each option. The right inverse gives back your input.

Inverse on finite sets (Seminar Task 7)

A = {2, 4, 6, 8, 10}, B = {1, 3, 5, 7, 9}, f(x) = x/2 + 1. (Assuming the garbled formula in the seminar sheet is x/2 + 1.)

x	2	4	6	8	10
f(x)	2	3	4	5	6
In B?	No	Yes	No	Yes	No

So f only works as a function into B on the domain **{4, 8}**, giving outputs **{3, 5}**. The inverse is $f^{-1}(x) = 2(x - 1) = 2x - 2$, mapping 3 → 4 and 5 → 8. The function and its inverse swap their input and output sets.

Graphs of inverses

The graph of f^{-1} is the **reflection** (mirror image) of the graph of f in the line **y = x**. Every point (a, b) on f becomes (b, a) on f^{-1}.

Mini tasks 7

1. Find the inverse of $f(x) = 5x - 7$.
2. Find the inverse of $f(x) = (x + 1)/4$.
3. Find the inverse of $f(x) = x^2 - 3$ for $x \geqslant 0$.
4. Which of these has no inverse over all real numbers: $2x + 1$, x^3, x^2, e^x?
5. If f(3) = 10, what is $f^{-1}(10)$?

8. Intersection of two functions

Two graphs **intersect** where they meet: the same x gives the same y in both.

Method:

1. Set them equal: f(x) = g(x).
2. Solve for x. (With a quadratic: move everything to one side, factorise, use the zero rule.)
3. Put each x into **either** function to get y. The simpler one (usually the line) is easier.
4. Write the points as (x, y).

Example 1 (lecture): $f(x) = x + 2$ and $g(x) = x^2$

- $x + 2 = x^2$
- $0 = x^2 - x - 2 = (x - 2)(x + 1)$
- $x = 2$ or $x = -1$
- $f(2) = 4$ and $f(-1) = 1$
- Intersection points: **(2, 4) and (−1, 1)**

Example 2 (Seminar Task 5): $p(x) = x^2 - 4x - 5$ and $q(x) = x + 1$

- $x^2 - 4x - 5 = x + 1$
- $x^2 - 5x - 6 = 0$
- $(x - 6)(x + 1) = 0$, so $x = 6$ or $x = -1$
- $q(6) = 7$ and $q(-1) = 0$
- Points: **(6, 7) and (−1, 0)**

Example 3 (revision Task 8): $v(x) = x^2 - 4x + 3$ and $w(x) = 2x - 5$

- $x^2 - 4x + 3 = 2x - 5 \rightarrow x^2 - 6x + 8 = 0 \rightarrow (x - 4)(x - 2) = 0$
- $w(4) = 3$, $w(2) = -1$
- Points: **(4, 3) and (2, −1)**

Example 4 (two lines): $y = 3x - 2$ and $y = -x + 6$

- $3x - 2 = -x + 6 \rightarrow 4x = 8 \rightarrow x = 2$. $y = 3(2) - 2 = 4$.
- One point: **(2, 4)**. Two different non-parallel lines always meet exactly once; parallel lines never meet.

Check: put your x into **both** functions. They must give the same y.

Mini tasks 8

1. Find where $y = x^2$ and $y = 2x + 3$ meet.
2. Find where $y = 2x + 1$ and $y = -x + 7$ meet.
3. Do $y = 3x + 1$ and $y = 3x - 2$ meet? Why or why not?

9. Recursive functions and the factorial

What recursion is

A **recursive function** is defined **in terms of itself**. To get a value, you use earlier values of the same function. It's like a staircase: to reach step 5, you first need step 4, which needs step 3, and so on, down to the ground floor.

Every recursive definition has two parts:

- **Base case (BC):** the starting value(s), given directly. This is the ground floor. Without it, the calculation would never end.
- **Recursive case (RC):** the rule that builds the next value from previous ones.

The Fibonacci sequence from Phase 1 is recursive: $F(n) = F(n-1) + F(n-2)$, with $F(0) = 0$, $F(1) = 1$.

The method: build a table from the bottom up

1. Write down the base case values.
2. Use the rule to get the next value. Write it down.
3. Repeat until you reach the value asked for.
4. **Write every step.** Most mistakes come from skipping steps or using the wrong earlier value.

Two styles of rule — read carefully

- **"f(n + 1) = … f(n) …"** builds the next value from the current one. To find f(1), put n = 0. To find f(2), put n = 1.
- **"f(n) = … f(n − 1), f(n − 2) …"** gives f(n) from the one or two before it. To find f(2), put n = 2.

When the rule contains **n itself** (like + n^2 or × n!), be careful which n you use. It is the n in that line of working.

Example 1 (lecture): BC $f(0) = 3$; RC $f(n + 1) = 2f(n) + 3$

n	Working	Value
0	Base case	f(0) = 3

n	Working	Value
1	2f(0) + 3 = 2(3) + 3	9
2	2f(1) + 3 = 2(9) + 3	21
3	2(21) + 3	45
4	2(45) + 3	93

Example 2 (mock Question 19): $g(0) = 1, g(1) = 2; g(n) = 3g(n-1) - g(n-2) + n^2$ for $n \geqslant 2$

n	Working	Value
0	Base case	1
1	Base case	2
2	$3g(1) - g(0) + 2^2 = 3(2) - 1 + 4$	9
3	$3g(2) - g(1) + 3^2 = 3(9) - 2 + 9$	34
4	$3g(3) - g(2) + 4^2 = 3(34) - 9 + 16$	**109**

Answer C (109). Note 34 is also an option (A): it's g(3), a trap for stopping one step early.

Example 3 (revision Task 7A): $s(0) = 2, s(1) = 4; s(n) = 2s(n-2) + s(n-1)$

s(2) = 2(2) + 4 = 8. s(3) = 2(4) + 8 = 16. s(4) = 2(8) + 16 = 32. s(5) = 2(16) + 32 = 64. s(6) = 2(32) + 64 = **128**.

Example 4 (Seminar Task 9.2): $t(0) = 1, t(1) = 3; t(n) = t(n-2) + t(n-1)$

1, 3, 4, 7, 11, 18, 29, 47. So **t(7) = 47**. (Count carefully: t(0) is the 1st value listed, t(7) is the 8th.)

Example 5 (Seminar Task 9.1): $p(0) = 2; p(n+1) = p(n)^2 + 1$

p(1) = 2^2 + 1 = 5. p(2) = 5^2 + 1 = 26. p(3) = 26^2 + 1 = 677. p(4) = 677^2 + 1 = 458329 + 1 = **458330**.

The factorial

n! ("n factorial") is the product of all whole numbers from n down to 1. It is a classic recursive function:

- **Base case:** 0! = 1
- **Recursive step:** n! = n × (n − 1)!

n	0	1	2	3	4	5	6
n!	1	1	2	6	24	120	720

So 5! = 5 × 4! = 5 × 24 = 120, which is 5 × 4 × 3 × 2 × 1.

Useful trick: 5!/3! = (5 × 4 × 3!)/3! = 5 × 4 = 20. Cancel instead of multiplying everything out.

Recursion with factorials

Seminar Task 9.3: f(0) = 1; f(n + 1) = f(n) + n!

Find	Use n =	Working	Value
f(1)	0	f(0) + 0! = 1 + 1	2
f(2)	1	f(1) + 1! = 2 + 1	3
f(3)	2	f(2) + 2! = 3 + 2	5
f(4)	3	f(3) + 3! = 5 + 6	**11**

The trap: to find f(4) you use **n = 3**, so you add 3!, not 4!.

Revision Task 7B: f(0) = 1; f(n + 1) = f(n) + (n + 1) × n!

Find	Use n =	Working	Value
f(1)	0	1 + 1 × 0! = 1 + 1	2
f(2)	1	2 + 2 × 1! = 2 + 2	4
f(3)	2	4 + 3 × 2! = 4 + 6	10
f(4)	3	10 + 4 × 3! = 10 + 24	**34**

Mini tasks 9

1. a(0) = 2; a(n) = a(n − 1) + 3n for n ⩾ 1. Find a(4).
2. b(0) = 1, b(1) = 1; b(n) = 2b(n − 1) + b(n − 2) + n for n ⩾ 2. Find b(4).
3. Work out 6!/4!.
4. c(0) = 0; c(n + 1) = c(n) + 2n + 1. Find c(1) to c(5). What pattern do you notice?

Phase 3 practice questions (exam style)

About 35 minutes, no calculator.

1. A = {1, 2, 3}, B = {x, y}. Which is a function with domain A and co-domain B? (A) {(1, x), (2, x)} (B) {(1, x), (2, y), (3, x), (2, x)} (C) {(1, y), (2, y), (3, y)} (D) {(1, x), (2, z), (3, y)}
2. Binary search for 23 in the sorted list 2, 5, 8, 12, 16, 23, 38, 56, 72, 91. Which elements are compared, in order? (A) 16, 56, 23 (B) 16, 38, 23 (C) 12, 38, 23 (D) 16, 72, 23
3. What is the list after the **first pass** of bubble sort on 7, 3, 9, 1, 5? (A) 3, 7, 1, 5, 9 (B) 1, 3, 5, 7, 9 (C) 3, 7, 9, 1, 5 (D) 3, 1, 7, 5, 9
4. What is the slope of the line through (−2, 5) and (4, −7)? (A) 2 (B) −2 (C) −1/2 (D) −6
5. What is the equation of the line through (2, −1) and (6, 7)? (A) y = 2x − 5 (B) y = 2x + 3 (C) y = ½x − 2 (D) y = 2x − 1
6. Where does y = −3x + 9 cross the x-axis? (A) (0, 9) (B) (−3, 0) (C) (3, 0) (D) (9, 0)
7. What is the slope of a line perpendicular to y = −(2/3)x + 1? (A) 2/3 (B) −3/2 (C) 3/2 (D) −2/3
8. Find the line through (3, 1) that is perpendicular to the line through (0, 0) and (2, 6). (A) y = 3x − 8 (B) y = −⅓x + 2 (C) y = ⅓x (D) y = −⅓x + 1
9. The lines y = 2(x − 3) + 5 and y = 2x + 1 are: (A) parallel (B) perpendicular (C) the same line (D) neither
10. What is the inverse of f(x) = (2x − 5)/3? (A) (3x − 5)/2 (B) (3x + 5)/2 (C) 3/(2x − 5) (D) (2x + 5)/3
11. What is the inverse of f(x) = 4 − x?
12. What is the inverse of f(x) = x² + 5 for x ⩾ 0? (A) √(x + 5) (B) √x − 5 (C) √(x − 5) (D) (x − 5)²
13. Why does f(x) = x² with domain ℝ have no inverse? (A) It is not onto (B) It is not one-to-one (C) It is not a function (D) It has no y-intercept
14. Where do y = x² − 2x and y = x + 4 intersect? (A) (4, 8) and (−1, 3) (B) (−4, 0) and (1, 5) (C) (4, 8) only (D) (2, 6) and (−2, 2)

15. Where do $y = 4x - 3$ and $y = -2x + 9$ intersect?
16. Describe the graph of $y = -2x + 4$: does it rise or fall, and where does it cross each axis?
17. $a(0) = 3$; $a(n + 1) = 2a(n) - 1$. What is $a(4)$? (A) 17 (B) 33 (C) 65 (D) 31
18. $h(0) = 1$, $h(1) = 1$; $h(n) = h(n - 1) + 2h(n - 2) + n$ for $n \geq 2$. What is $h(4)$? (A) 10 (B) 24 (C) 20 (D) 29
19. $f(0) = 2$; $f(n + 1) = f(n) + (n + 1)!$. What is $f(4)$? (A) 35 (B) 11 (C) 26 (D) 59
20. What is $7!/5!$? (A) 2 (B) 42 (C) 35 (D) 7/5
21. $g(x) = (3x + 2) \bmod 5$. What is $g(6)$? (A) 20 (B) 4 (C) 0 (D) 2
22. $A = \{1, 2, 3\}$. Which function from A to A is **not** one-to-one? (A) $\{(1, 2), (2, 3), (3, 1)\}$ (B) $\{(1, 1), (2, 2), (3, 3)\}$ (C) $\{(1, 1), (2, 1), (3, 2)\}$ (D) $\{(1, 3), (2, 2), (3, 1)\}$

Answers with full explanations

Mini tasks 1 (algorithms)

1. **4 comparisons:** 12, 34, 25, then 6 matches.
2. L = 1, R = 9: M = 5, $a_m = 27 < 43$, so L = 6. M = $\lfloor 15/2 \rfloor = 7$, $a_m = 37 < 43$, so L = 8. M = $\lfloor 17/2 \rfloor = 8$, $a_m = 43$. **Found at position 8 after 3 comparisons.**
3. (5, 1) swap → 1, 5, 4, 2, 8. (5, 4) swap → 1, 4, 5, 2, 8. (5, 2) swap → 1, 4, 2, 5, 8. (5, 8) no swap. **1, 4, 2, 5, 8.**
4. The list is **not sorted**. Binary search relies on order to know which half to throw away.

Mini tasks 3 (functions)

1. **Function.** Each of 1, 2, 3 has one output.
2. **Not a function.** 3 has no output.
3. **Not a function.** 1 has two outputs.
4. **Function.** All going to y is allowed.
5. $f(0) = -1$, $f(1) = 1$, $f(2) = 3$. **Range $\{-1, 1, 3\}$.**

Mini tasks 5 (lines)

1. m = **−3**, b = **7**.
2. $(-4 - 4)/(3 - (-1)) = -8/4 =$ **−2**.
3. $m = (7 - 3)/(3 - 1) = 2$. $3 = 2(1) + b$, so $b = 1$. **$y = 2x + 1$.**

4. y-intercept: **(0, −2.5)**. x-intercept: (3/4)x = 5/2, so x = 10/3 ≈ 3.33: **(10/3, 0)**.
5. m = −1 < 0, so it **falls**. It crosses the y-axis at **(0, 2)**.

Mini tasks 6 (parallel and perpendicular)

1. (−3) × (1/3) = −1: **perpendicular**.
2. Flip 2/7 to 7/2 and change sign: **−7/2**.
3. Slope −1/4. 2 = (−1/4)(4) + b = −1 + b, so b = 3. **y = −¼x + 3.**
4. Slope 5. 1 = 5(1) + b, so b = −4. **y = 5x − 4.**

Mini tasks 7 (inverses)

1. x = 5y − 7 → x + 7 = 5y → **$f^{-1}(x) = (x + 7)/5$**.
2. x = (y + 1)/4 → 4x = y + 1 → **$f^{-1}(x) = 4x − 1$**.
3. x = y^2 − 3 → y^2 = x + 3 → **$f^{-1}(x) = \sqrt{(x + 3)}$**.
4. **x^2.** It's not one-to-one: 2 and −2 give the same output. (x^3 is one-to-one because cubes keep the sign.)
5. **3.** The inverse sends the output back to its input.

Mini tasks 8 (intersections)

1. x^2 = 2x + 3 → x^2 − 2x − 3 = 0 → (x − 3)(x + 1) = 0. **(3, 9) and (−1, 1).**
2. 2x + 1 = −x + 7 → 3x = 6 → x = 2, y = 5. **(2, 5).**
3. **No.** Both have slope 3 with different intercepts, so they're parallel. (Setting them equal gives 1 = −2, which is impossible.)

Mini tasks 9 (recursion)

1. a(1) = 2 + 3 = 5; a(2) = 5 + 6 = 11; a(3) = 11 + 9 = 20; a(4) = 20 + 12 = **32**.
2. b(2) = 2(1) + 1 + 2 = 5; b(3) = 2(5) + 1 + 3 = 14; b(4) = 2(14) + 5 + 4 = **37**.
3. 6!/4! = 6 × 5 = **30**.
4. c(1) = 0 + 0 + 1 = 1; c(2) = 1 + 2 + 1 = 4; c(3) = 4 + 4 + 1 = 9; c(4) = 9 + 6 + 1 = 16; c(5) = 16 + 8 + 1 = 25. **They are the square numbers: $c(n) = n^2$.**

Phase 3 practice questions

1. **(C)** Each input has exactly one output in B. (A) leaves out 3. (B) gives 2 two outputs. (D) uses z, which isn't in B.

2. **(A)** L = 1, R = 10: M = 5 → 16 < 23, so L = 6. M = ⌊16/2⌋ = 8 → 56 > 23, so R = 7. M = ⌊13/2⌋ = 6 → 23. Found.
3. **(A)** (7, 3) swap → 3, 7, 9, 1, 5. (7, 9) no swap. (9, 1) swap → 3, 7, 1, 9, 5. (9, 5) swap → 3, 7, 1, 5, 9. (B) is the fully sorted list, not the first pass.
4. **(B)** (−7 − 5)/(4 − (−2)) = −12/6 = −2.
5. **(A)** m = (7 − (−1))/(6 − 2) = 8/4 = 2. −1 = 2(2) + b, so b = −5.
6. **(C)** 0 = −3x + 9, so x = 3. (A) is the y-intercept, a trap.
7. **(C)** Flip 2/3 to 3/2 and change the sign from − to +.
8. **(B)** The slope through (0, 0) and (2, 6) is 3. Perpendicular slope −1/3. 1 = (−1/3)(3) + b = −1 + b, so b = 2.
9. **(A)** Expand: 2x − 6 + 5 = 2x − 1. Same slope 2, different intercepts (−1 versus 1).
10. **(B)** x = (2y − 5)/3 → 3x = 2y − 5 → 3x + 5 = 2y → y = (3x + 5)/2. Quick check: f(4) = 1, and (3 × 1 + 5)/2 = 4. ✓
11. **$f^{-1}(x) = 4 - x$.** x = 4 − y → y = 4 − x. This function is its own inverse. Check: f(1) = 3 and f(3) = 1.
12. **(C)** x = y² + 5 → y² = x − 5 → y = √(x − 5).
13. **(B)** f(2) = f(−2) = 4, so going backwards from 4 is ambiguous.
14. **(A)** x² − 2x = x + 4 → x² − 3x − 4 = 0 → (x − 4)(x + 1) = 0. x = 4 gives y = 8; x = −1 gives y = 3.
15. **(2, 5).** 4x − 3 = −2x + 9 → 6x = 12 → x = 2, y = 4(2) − 3 = 5.
16. **Falls** (slope −2). Crosses the y-axis at **(0, 4)** and the x-axis at **(2, 0)** (0 = −2x + 4).
17. **(B)** a(1) = 5, a(2) = 9, a(3) = 17, a(4) = 33. (A) is a(3), the stop-early trap.
18. **(B)** h(2) = 1 + 2(1) + 2 = 5. h(3) = 5 + 2(1) + 3 = 10. h(4) = 10 + 2(5) + 4 = 24.
19. **(A)** f(1) = 2 + 1! = 3. f(2) = 3 + 2! = 5. f(3) = 5 + 3! = 11. f(4) = 11 + 4! = 35. (To find f(4) use n = 3, so add (3 + 1)! = 4!.)
20. **(B)** 7 × 6 = 42.
21. **(C)** 3(6) + 2 = 20, and 20 mod 5 = 0.
22. **(C)** 1 and 2 both go to 1, so two inputs share an output.

Scoring: 18 or more out of 22 is strong. If you missed recursion questions, redo them with a table. If you missed line questions, re-read the slope and perpendicular sections.

When you're ready, ask for **Phase 4: Logic**.